

Historical and Source Companion to RGCS v3 / RSCS 1.0

Andrew Green

2026

Abstract

This companion preserves the source material behind the RGCS programme — historical practice, personal logs, theoretical proposals, and recent peer-reviewed photonics — in its own words and with its own authors, and records exactly what the programme did with each piece: what mathematics was adapted (always re-derived and re-tested), what interpretation was *excluded* by a binding forbidden-transfer clause, and which testable hypotheses resulted. Nothing here is evidence for anything; that is the point of keeping it separate. The register covers 9 sources and 19 equation-level adaptation rows.

1 Why a companion

Programmes that draw inspiration from unconventional sources face two symmetric failure modes: laundering (source claims quietly become physics by repetition) and erasure (the historical material is discarded or ridiculed, destroying the record of where questions came from). The RGCS discipline avoids both with one rule: **every statement keeps its author and its class**. A source claim [SRC] may inspire a hypothesis [HYP] only through an explicit, registered translation with an observable, controls, and a failure condition — and the original wording is preserved here unchanged.

2 The historical crystal-practice corpus [SRC]

The programme’s origin material is a practice tradition around faceted quartz instruments (the Vogel-style cut lineage) and a personal working log (“JH log”) whose operating points parameterize the v2/v3 protocols: the 4096 Hz coil carrier; the sound keys at 1496, 644, and 587 Hz; the 20 Hz electrode-pulse family with its 15 V setting; the 46 ms burst envelopes with their spacings and pauses; opposed complementary coils; and “the eye” — a special interior locus of the crystal. In the source’s own frame these are statements of practice, made in earnest, with their own vocabulary of resonance, keys, and awakening. The programme records them respectfully and verbatim in the source-evidence ledger, assigns each an RG-* identifier, and gives them *no truth value*: they select where to look, never what is true.

Two translations illustrate the method. “A single crystal tone” becomes the scalar effective wave speed v_L RGCS-M.10 — which v3’s anisotropic model then resolves, recovering the scalar as the orientation-unknown case. “The eye” becomes an eight-entry node menu (metric centre, shaft prior, measured vibration node, electrical node, optical feature, overlap maximum, coupling maximum, phase singularity), of which the measured node supersedes all others and five entries are pre-registered hypotheses H-24 – H-28 with failure conditions.

3 The Arisaka NHT/HAL proposals [SRC]

The Neural Holography Theory and Holographic Ring Attractor Lattice proposals [1–3] are theoretical neuroscience: space encoded in the phase of oriented traveling waves, ring-attractor lattices as memory, egocentric ↔ allocentric transforms. They are their authors’ proposals and remain so. The programme adapted their *structure* — a typed memory record with allocentric/egocentric positions, a frame transform, predicted vs. observed states RSCS-C.15, [ENG] — which is useful software regardless of the neuroscience, while the space-to-phase encoding RSCS-O.3 and memory lattice RSCS-C.14 are quarantined [HYP] requiring an explicit acknowledgment flag to construct. **Excluded outright**: any consciousness claim, any brain-mechanism claim, and any claim that quartz realizes memory.

4 The photonics source set [\[SRC\]](#) → adapted mathematics

Six recent works supplied the optical/nonreciprocal mathematics: phonon-mediated Autler–Townes splitting [4], terahertz-bandwidth acousto-optic isolation [5], self-induced nonlinear nonreciprocity [6], noiseless single-photon isolation [7], TMOKE coupled-waveguide beating [8], and latched-garnet Faraday isolation via spiral-helix laser writing [9]. Each achieves its effect through an active ingredient quartz is not claimed to have; the binding posture is that an unbiased passive quartz path is *reciprocal* and every directional observable’s pre-registered expectation is the null. The complete equation-level ledger — what crossed, what is forbidden — follows.

Table 1: Equation-level adaptation ledger: every reused piece of source mathematics with its binding forbidden transfer (references/equation_provenance.yaml, frozen).

EP id	Source	Adapted mathematics	Forbidden transfer (binding)
EP-01-01	SRC-3-01	Two-mode coupled equations of motion (WGR + waveguide), driven	no claim that quartz hosts acousto-optic ATS, WGR modes, or the reported 39 dB isolation; device numbers stay with the paper
EP-01-02	SRC-3-01	Autler-Townes dressed-state splitting of the cavity susceptibility	the physical splitting is optical; not evidence for a quartz acoustic ATS effect
EP-01-03	SRC-3-01	Critical-coupling transmission zero	device insertion-loss/contrast values are the paper's
EP-02-01	SRC-3-02	Acousto-optic phase-matching condition and phase mismatch	no claim quartz realizes THz-bandwidth optical isolation or CMOS-photonics beam splitters
EP-02-02	SRC-3-02	AOM-section transfer matrix (parity basis)	acoustic field b , group velocities $v_{\pm}$, and device geometry stay optical-domain
EP-02-03	SRC-3-02	Delay-region matrix and forward/backward interference (group-delay balancing)	no import of nonreciprocity/isolation as an intrinsic quartz property
EP-03-01	SRC-3-03	Alternating spiral-helix filament laser-writing path	latched-magnetization / Faraday-rotation physics is BIG-specific; NOT a quartz effect. Preprint (not peer-reviewed) – SRC weight.
EP-03-02	SRC-3-03	Crossed-polarizer Faraday isolator assembly (45-degree offset)	no quartz Faraday rotation claim; figures of merit are the paper's
EP-04-01	SRC-3-04	Anti-symmetric (gyrotropic) susceptibility tensor	the medium is atomic vapor; no claim quartz has a self-induced χ_{xy}
EP-04-02	SRC-3-04	Nonlinear (signal-induced) non-reciprocal susceptibility expansion	no import of the physical claim that a signal self-induces nonreciprocity in quartz
EP-04-03	SRC-3-04	Nonreciprocal transmittance, phase, and isolation	atomic-vapor performance stays the paper's
EP-05-01	SRC-3-05	Velocity-selective optical pumping (Doppler) coordinate	no claim quartz has Doppler-selected atomic populations; single-photon/quantum regime not imported
EP-05-02	SRC-3-05	Insertion loss / isolation and single-photon fidelity metrics	$g^{(2)}$ single-photon statistics are quantum-optical; not claimed for classical quartz acoustics
EP-06-01	SRC-3-06	Nonreciprocal propagation-constant shift (TMOKE)	the perturbation is magneto-optic (Kerr); no quartz analogue asserted
EP-06-02	SRC-3-06	Even/odd supermode decomposition of a coupled system	waveguide-specific field profiles are the paper's

EP id	Source	Adapted mathematics	Forbidden transfer (binding)
EP-06-03	SRC-3-06	Modal beating length and transfer-matrix cascade	device isolation performance is the paper's
EP-07-01	SRC-3-07	NHT space-to-phase (holographic plane-wave) representation	NO consciousness/brain/memory conclusion may be imported into any RGCS physical claim; NHT is not evidence for anything about quartz
EP-07-02	SRC-3-07	Holographic Ring Attractor Lattice (HAL) memory-lattice coordinate	no brain-memory mechanism asserted as physically realized in quartz or any RGCS device
EP-08-01	SRC-3-08	Scaling/rotation multidimensional transformations (vision)	no visual-perception conclusion imported into RGCS physics

5 Evidence boundaries, stated plainly

As of this writing: no experiment in this programme has confirmed any non-standard physical hypothesis. The v2 measurement campaigns produced calibrations, controls, artifact identifications, and negative results — all registered. The value the historical material has contributed is real but methodological: it generated precise, falsifiable questions that conventional instrumentation can answer. If every pre-registered hypothesis fails, the companion still stands as an honest record of a tradition, its language, and a disciplined attempt to test it.

References

- [1] Katsushi Arisaka. Grand unified theory of mind and brain, part II: Neural holographic tomography (NHT) and holographic ring attractor lattice (HAL). V1 manuscript, 2022-03-05. University of California, Los Angeles. SRC-class source; not peer-reviewed, 2022.
- [2] Katsushi Arisaka and Aaron Blaisdell. Grand unified theory of mind and brain, part III: Holographic visual perception of 3D space and shapes. V1 manuscript, 2022-03-05. University of California, Los Angeles. SRC-class source; not peer-reviewed, 2022.
- [3] Katsushi Arisaka. UCLA physics colloquium presentation, 2023-05-25. Slide deck; presentation context for arisaka2022nht/arisaka2022vision. SRC-class source, 2023.
- [4] Donggyu B. Sohn, Oğulcan E. Örsel, and Gaurav Bahl. Electrically driven linear optical isolation through phonon mediated Autler-Townes splitting, 2021. Department of Mechanical Science and Engineering, University of Illinois at Urbana–Champaign.
- [5] Haotian Cheng, Yishu Zhou, Freek Ruesink, Margaret Pavlovich, Shai Gertler, Andrew L. Starbuck, Andrew J. Leenheer, Andrew T. Pomerene, Douglas C. Trotter, Christina Dallo, Matthew Boady, Katherine M. Musick, Michael Gehl, Ashok Kodigala, Matt Eichenfield, Anthony L. Lentine, Nils T. Otterstrom, and Peter T. Rakich. A terahertz bandwidth nonmagnetic isolator, 2024.
- [6] Zhu-Bo Wang, Yan-Lei Zhang, Xin-Xin Hu, Guang-Jie Chen, Ming Li, Peng-Fei Yang, Xu-Bo Zou, Peng-Fei Zhang, Chun-Hua Dong, Gang Li, Tian-Cai Zhang, Guang-Can Guo, and Chang-Ling Zou. Self-induced optical non-reciprocity. *Light: Science & Applications*, 14:23, 2025. doi: 10.1038/s41377-024-01692-y.
- [7] Shicheng Zhang, Yifan Zhan, Shangqing Gong, and Yueping Niu. Noiseless single-photon isolator at room temperature. *Communications Physics*, 2023. doi: 10.1038/s42005-023-01141-z.
- [8] Kimhong Chao, Vy Yam, Laurent Vivien, and Béatrice Dagens. Integrated TE optical isolator based on magneto-optical perturbation in coupled waveguides. *Scientific Reports*, 2025. doi: 10.1038/s41598-025-08507-z.
- [9] Jerome Lapointe, Cedrik Coia, and Réal Vallée. Integrated magnetless passive broadband Faraday isolator, 2023. Research Square preprint, posted 2023-07-25; not peer-reviewed.